RML stability — benchmark report

Agent-alone candidates, published prediction values and reference variants calculated in this work.

Agent Scientist: gpt-6.1-sol. Predictor names and units are listed with each measurement.

Published predictions and our predictions

Property / predictor	Published prediction	Our candidates
Vina docking score, pNPL (kcal/mol)	WT −5.5; N120M/E230I/N264M −5.6	RML_R2_01 −5.221; RML_R2_02 −5.193
Folding-stability predictions	FoldX: N120M −0.35775; E230I −1.59876; N264M −0.77627 kcal/mol (single mutants). Additional Table 1 predictors in references.	DDGun_SEQ: RML_R2_02 −0.1; RML_R2_01 −0.2 kcal/mol ΔΔGunfold
Predicted Tm change (°C)	Not reported	RML_R2_02 +1.0722; RML_R2_01 +1.0096

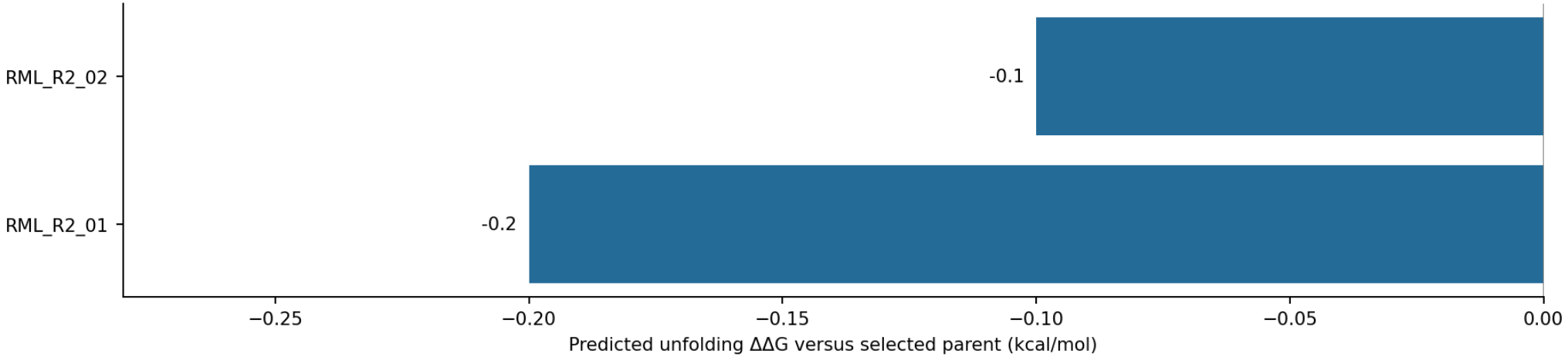
Source: Teng et al., Tables 1 and sections 3.4/3.7

Our stability predictions

Candidate	Mutations	DDGun unfolding ΔΔG (kcal/mol)
RML_R2_02	S11Q,S71K,E72D,S135D,R160L,E161K,E162Q	-0.1
RML_R2_01	S11Q,S71K,E72T,S135D,R160L,E161K,E162Q	-0.2

Change is relative to our selected parent. Positive DDGun unfolding ΔΔG indicates predicted stabilization; negative indicates predicted destabilization.

Stability change from our parent

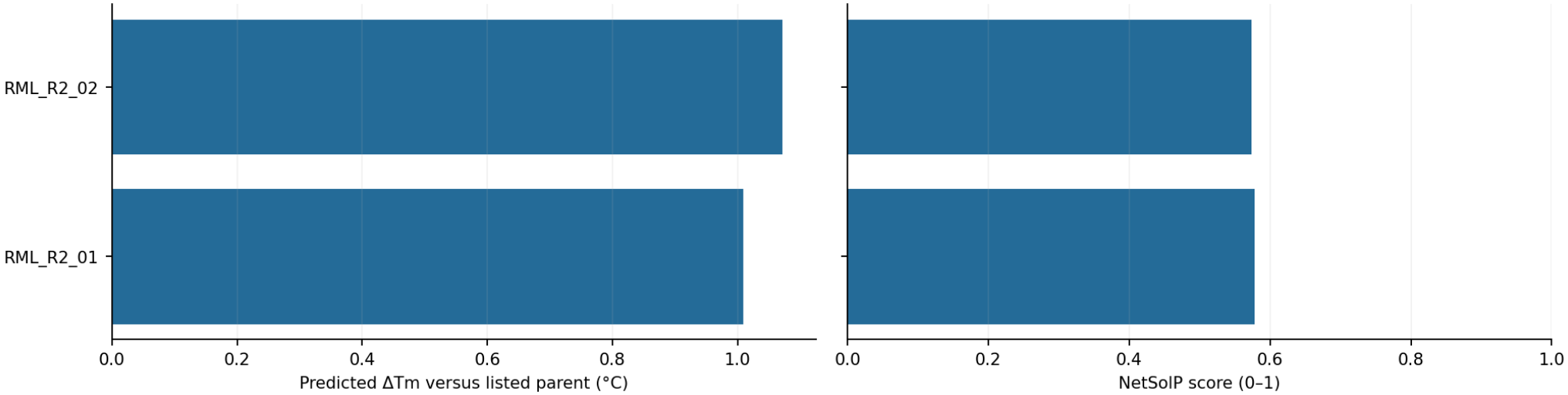


All bars use DDGun_SEQ and the same selected parent within this task.

Property predictions calculated in this work

Sequence	Group	Parent / background	Predicted Tm (°C)	ΔTm (°C)	NetSolP	pHPred (raw)
RML_R2_02	Our candidate	RML_WT	48.0504	1.0722	0.573641	6.9071
RML_R2_01	Our candidate	RML_WT	47.9878	1.0096	0.577789	6.8929

Candidate and reference property predictions



Blue: our candidates. Amber: reference variants calculated in this work. Changes use the parents listed in the table.

Parent identity, preparation details and run records: results/task_001_rml_stability/agent-alone/README.md.